

CHIMBAY, Uzbekistan

WMO: 382620

Lat: 42.950N

Lon: 59.800E

Elev: 93

StdP: 100.21

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.4	-16.3	-23.6	0.5	-18.7	-20.8	0.6	-15.7	9.1	-1.8	8.1	-2.2	2.5	50	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	38.7	22.3	37.1	21.5	35.6	20.8	24.2	35.3	23.0	34.1	22.0	33.4	2.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.6	30.8	19.1	14.0	29.7	17.8	12.9	29.0	73.2	35.1	68.7	34.2	64.9	33.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.1	7.1	-20.9	41.9	4.2	1.8	-23.9	43.2	-26.3	44.3	-28.6	45.3	-31.7	46.6		
(5)			-20.9	26.1	3.8	1.8	-23.6	27.4	-25.8	28.4	-28.0	29.4	-30.7	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-4.3	-1.9	6.3	14.6	21.7	26.3	28.0	26.1	19.3	11.7	3.7	-2.1	(6)
(7)		DBStd	12.36	5.87	6.07	5.58	5.14	3.80	3.12	2.74	3.30	3.92	4.71	5.42	5.13	(7)
(8)		HDD10.0	1544	443	333	141	17	0	0	0	0	1	37	198	374	(8)
(9)		HDD18.3	3106	702	566	374	133	14	1	0	1	34	209	440	632	(9)
(10)		CDD10.0	2469	0	1	26	155	362	490	559	500	279	90	8	0	(10)
(11)		CDD18.3	989	0	0	1	21	118	241	301	242	63	4	0	0	(11)
(12)		CDH23.3	12771	0	0	18	287	1421	3187	4044	2986	742	83	3	0	(12)
(13)		CDH26.7	6794	0	0	4	98	624	1776	2351	1649	277	14	1	0	(13)
(14)	Wind	WSAvg	2.5	2.6	2.9	3.2	3.1	2.9	2.7	2.3	2.1	2.1	2.2	2.2	2.4	(14)
(15)	Precipitation	PrecAvg	123	12	11	18	19	17	5	2	3	3	9	11	12	(15)
(16)		PrecMax	254	41	53	55	84	72	31	26	15	21	38	45	51	(16)
(17)		PrecMin	61	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	45	8	12	14	18	18	6	4	4	4	8	12	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	16.2	26.0	32.9	36.3	39.8	41.4	40.2	35.4	28.5	21.0	12.3	(19)
(20)			MCWB	6.4	9.9	14.1	18.5	19.5	23.2	22.8	23.3	20.8	15.8	12.6	7.9	(20)
(21)		2%	DB	7.2	11.4	21.1	29.7	34.0	37.7	39.0	37.7	32.1	25.6	16.9	8.8	(21)
(22)			MCWB	4.2	6.7	12.4	16.9	19.4	21.7	22.2	22.2	19.2	14.9	10.6	5.1	(22)
(23)		5%	DB	4.6	8.3	18.0	26.7	32.2	36.1	37.2	36.0	29.8	23.2	14.2	6.5	(23)
(24)			MCWB	2.4	4.5	10.4	15.1	18.6	20.8	21.5	21.4	17.8	13.8	8.9	3.5	(24)
(25)		10%	DB	2.8	5.8	15.1	23.9	30.3	34.5	35.6	34.2	27.6	20.5	11.6	4.6	(25)
(26)			MCWB	1.0	2.8	8.9	13.8	17.7	20.0	21.0	20.6	16.7	12.4	7.4	2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	10.3	14.7	19.7	22.9	25.5	25.4	26.6	21.9	17.3	13.8	8.2	(27)
(28)			MCD B	10.5	15.8	24.7	30.2	33.3	35.8	36.2	36.9	32.0	26.3	19.2	11.6	(28)
(29)		2%	WB	4.3	7.0	12.7	17.8	20.7	23.6	23.8	24.0	20.1	15.6	11.4	5.5	(29)
(30)			MCD B	6.7	10.6	20.2	27.8	31.2	35.0	35.4	34.7	31.1	24.2	16.2	8.4	(30)
(31)		5%	WB	2.6	4.9	11.0	16.1	19.4	22.3	22.9	22.5	18.6	14.1	9.4	3.7	(31)
(32)			MCD B	4.5	8.1	17.4	25.3	29.8	33.7	34.1	33.5	28.1	22.1	13.6	6.2	(32)
(33)		10%	WB	1.1	3.0	9.2	14.5	18.4	21.1	22.1	21.3	17.3	12.7	7.6	2.3	(33)
(34)			MCD B	2.8	5.6	14.6	22.5	28.5	32.0	33.5	32.4	26.2	19.9	11.3	4.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.7	11.6	12.8	13.5	13.8	13.7	14.4	14.6	13.9	10.5	8.2	(35)
(36)			MCDBR	9.8	13.2	16.1	16.5	16.4	16.3	16.1	16.6	16.8	17.4	14.4	10.8	(36)
(37)			MCWBR	7.7	9.3	9.2	7.5	6.9	6.4	6.0	6.8	7.9	9.3	9.2	8.3	(37)
(38)		5% WB	MCDBR	9.4	12.6	14.7	15.7	14.6	15.3	14.9	15.9	16.1	15.8	12.8	9.9	(38)
(39)			MCWBR	7.6	9.1	8.8	8.0	7.1	6.9	6.3	7.2	8.2	9.0	8.6	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.377	0.405	0.467	0.522	0.526	0.511	0.493	0.469	0.441	0.446	0.410	0.381	(40)	
(41)		taud	2.061	2.007	1.914	1.821	1.858	1.927	2.001	2.035	2.050	1.983	2.036	2.058	(41)	
(42)		Ebn at Noon	707	758	759	751	762	776	785	789	777	702	660	657	(42)	
(43)		Edh at Noon	110	137	170	202	199	187	172	161	147	138	112	101	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.51	2.51	3.91	5.36	6.65	7.45	7.18	6.58	5.19	3.42	2.04	1.39	(44)	
(45)		RadStd	0.18	0.30	0.33	0.38	0.38	0.33	0.25	0.21	0.21	0.20	0.24	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A	N/A	+53	
(47) Regional (4 neighbors)	N/A	N/A	N/A	+0.83	N/A	-0.67	N/A	N/A	N/A	+52	

Nomenclature: See separate page